Spectral Contraction and T-Independence in Barrington Branching Programs: Resolving the Uniform Mixing Conjecture for Point Functions over S_5

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Abstract

We study the distribution D_T over the symmetric group S_5 induced by evaluating a Barrington branching program for the point function CC_T on uniformly random inconsistent inputs. We prove two results. First, $D_T = D_{T'}$ for all $T, T' \in \{0,1\}^\ell$ (exact T-independence), via a combinatorial flip bijection that requires no properties of the underlying group. Second, $d_{\mathrm{TV}}(D_T, \mathrm{Uniform}(S_5)) \leq \mathrm{negl}(\ell)$ (spectral contraction), via Fourier analysis on S_5 using the Diaconis–Shahshahani Upper Bound Lemma. The contraction proof proceeds by exact computation of Fourier norms for the standard representation $(4,1)$ at the first two levels of the commutator tree, followed by exhaustive verification that the operator norm drops below 1 at the third level — enabled by the discovery that the level-2 obstruction has a regular simplex structure in $\mathbb{R}^4$ that the commutator necessarily breaks. Together, these results resolve the Uniform T-Independence Conjecture for Barrington point-function branching programs and establish an information-theoretic foundation for the security of branching-program-based obfuscation of point functions and conjunctions.

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1 Introduction

Indistinguishability obfuscation (iO) for general circuits is one of the most powerful primitives in cryptography, implying a vast array of applications from functional encryption to deniable encryption [10, 18]. The first candidate constructions [10] were based on matrix branching programs over multilinear maps, randomized via Kilian's technique [15], following the classical correspondence established by Barrington's theorem [1]: every NC^1 circuit of depth d can be simulated by a branching program of width 5 and length at most 4^d over the symmetric group S_5.

The security of such constructions depends critically on the algebraic properties of the distributions that arise when branching programs are evaluated on "incorrect" inputs. Specifically, for a point function $\mathrm{CC}_T : \{0,1\}^\ell \to \{0,1\}$ that outputs 1 only on $x = T$, the Barrington program BP_T produces a designated permutation $\pi_{\text{accept}} \in S_5$ on input T and the identity e on all other inputs. When the program is evaluated on a random bit string $b \in \{0,1\}^L$ that is *inconsistent* (at least one variable receives different values at different steps), the output is a random element of S_5 whose distribution we denote D_T.

Two properties of D_T are essential for security:

(i) **T-independence:** $D_T = D_{T'}$ for all T, T', so the distribution reveals nothing about which target was chosen.

(ii) **Uniform mixing:** $d_{\mathrm{TV}}(D_T, \mathrm{Uniform}(S_5)) \leq \mathrm{negl}(\ell)$, so the distribution is indistinguishable from random.

These properties were widely assumed but not formally established: the distributional behavior of Barrington programs on inconsistent inputs — a foundational question open since 1989 — was implicitly relied upon in security analyses without proof. Property (i) was used without justification, and property (ii) was supported by heuristic arguments and limited computational evidence.

Our contributions. We prove both properties.

For T-independence, we give a short combinatorial proof (Theorem 3.1) via a flip bijection F_S that maps inconsistent paths for T to inconsistent paths for T' while preserving the output permutation. The proof uses no property of S_5 and extends to conjunctions with wildcards.

For uniform mixing, we develop a Fourier-analytic framework on S_5 (Section 4). Using the Diaconis–Shahshahani Upper Bound Lemma, we reduce the problem to bounding the Fourier norms of D_T in each irreducible representation. The standard representation $(4,1)$ of dimension 4 is the bottleneck. We establish contraction at the first two levels of the commutator tree by exact rational computation (Theorems 4.3 and 4.4), discover that the level-2 obstruction has a regular simplex structure (Theorem 4.5), and prove that this structure is necessarily broken at level 3 (Theorem 4.8), yielding operator-norm contraction with $\lambda \leq 0.0354$. An inductive argument then gives super-exponential decay for all higher levels (Theorem 4.9).

Implications. Combined, our results show that for $\ell \geq 8$, the distribution over inconsistent evaluations of BP_T is simultaneously independent of T and negligibly close to uniform. This provides an information-theoretic foundation for the security of obfuscation schemes based on Barrington branching programs for point functions and, more broadly, for conjunctions with bounded wildcard density.

Related work. The first iO candidate from branching programs [10] was followed by constructions for specific function classes: lockable obfuscation for point functions from LWE [11, 12], conjunction obfuscation from noisy encodings and LPN [5, 6], and entropic Ring-LWE approaches [4]. The landmark result of Jain, Lin, and Sahai [14] achieved iO for general circuits from well-founded assumptions, departing from the branching-program paradigm. Recent lattice-based candidates [3, 19, 13] use matrix groups $\mathrm{GL}_n(\mathbb{Z}_q)$ rather than S_5, while impossibility results [9] constrain the space of viable assumptions. For these constructions, the mixed-input security model — where an adversary evaluates the program on inconsistent inputs — remains central [8].

Fourier-analytic methods for branching programs have a separate tradition in the pseudorandomness literature [17, 16], but that work uses Fourier analysis on $\{0,1\}^n$ (the input space) to fool Boolean outputs of read-once programs. Our analysis instead uses Fourier analysis on S_5 (the output group) to bound total variation of the group-valued output distribution of multi-read programs — a fundamentally different analytic object requiring different tools. We are not aware of prior work connecting the Diaconis–Shahshahani framework [7] to branching-program security.

Techniques. The proof of T-independence is purely combinatorial. The proof of spectral contraction uses three different methods at three different scales: exact symbolic algebra (levels 0–2), exhaustive numerical verification (level 3), and analytic induction (levels 4+). The key discovery that bridges levels 2 and 3 is the simplex structure of the preserved directions: the 270 out of 405 level-2 configurations with operator norm 1 each preserve one of exactly 5 directions in $\mathbb{R}^4$, forming a regular simplex with pairwise $|\cos\theta| = 1/4$. The commutator structure at level 3 forces sub-blocks from different directions, breaking the obstruction.

2 Preliminaries

2.1 Branching Programs and Barrington's Theorem

Definition 2.1 (Permutation Branching Program)**.** *A* permutation branching program *of width w and length L over a group $G \leq S_w$ is a sequence of instructions $\Pi = ((i_1, g_1^{(0)}, g_1^{(1)}), \ldots, (i_L, g_L^{(0)}, g_L^{(1)}))$, where each $i_j \in [\ell]$ is a variable index and $g_j^{(0)}, g_j^{(1)} \in G$. On input $x \in \{0,1\}^\ell$, the program evaluates $\Pi(x) = \prod_{j=1}^{L} g_j^{(x_{i_j})}$. The program* α-computes *a function f if $\Pi(x) = \alpha$ when $f(x) = 1$ and $\Pi(x) = e$ when $f(x) = 0$.*

Theorem 2.2 (Barrington [1])**.** *Every Boolean function computed by a circuit of depth d can be α-computed by a permutation branching program of width 5 and length at most 4^d over S_5, for some $\alpha \neq e$.*

The construction uses the *commutator trick*: given programs P and Q that α-compute f and β-compute g respectively, the concatenation $P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ computes $f \wedge g$ via the group commutator $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$. This works because S_5 is non-solvable: iterated commutators of non-identity elements remain non-identity.

Definition 2.3 (Point Function)**.** *The point function $\mathrm{CC}_T : \{0,1\}^\ell \to \{0,1\}$ is defined by $\mathrm{CC}_T(x) = 1$ if and only if $x = T$.*

For point functions, the Barrington program has ℓ leaves (one per bit position), each comparing x_i to T_i. Leaf i outputs a fixed transposition σ_i when $x_i = T_i$ and the identity e when $x_i \neq T_i$. The tree of commutators has depth $\log_2 \ell$, yielding a program of length $L = 4^{\log_2 \ell} = \ell^2$.

2.2 The Distribution D_T

Definition 2.4 (Inconsistent Path)**.** *A bit string $b = (b_1, \ldots, b_L) \in \{0,1\}^L$ is* inconsistent *with respect to the program Π if there exist steps j, j' reading the same variable ($i_j = i_{j'}$) with $b_j \neq b_{j'}$.*

Definition 2.5 (D_T)**.** *Let $\mathcal{I} \subset \{0,1\}^L$ be the set of inconsistent paths. The distribution D_T over S_5 is defined by*

$$D_T(\pi) = \Pr_{b \sim \mathrm{Uniform}(\mathcal{I})} \left[\prod_{j=1}^{L} g_j^{(b_j)} = \pi \right].$$

2.3 Representation Theory of S_5

The symmetric group S_5 has 7 irreducible representations, labeled by partitions of 5:

Partition λ	Dimension d_λ	χ_λ(transposition)	Name
(5)	1	1	Trivial
$(4,1)$	4	2	Standard
$(3,2)$	5	1	—
$(3,1,1)$	6	0	—
$(2,2,1)$	5	-1	Sign $\otimes$ $(3,2)$
$(2,1,1,1)$	4	-2	Sign $\otimes$ Standard
(1^5)	1	-1	Sign

The dimensional check confirms: $1^2 + 4^2 + 5^2 + 6^2 + 5^2 + 4^2 + 1^2 = 120 = |S_5|$.

Definition 2.6 (Standard Representation)**.** *The standard representation $\rho_{(4,1)} : S_5 \to \mathrm{GL}(\mathbb{R}^4)$ acts on the hyperplane $V = \{v \in \mathbb{R}^5 : \sum_i v_i = 0\}$ by permuting coordinates. We use the orthonormal basis*

$$f_1 = (e_0 - e_1)/\sqrt{2}, \quad f_2 = (e_0 + e_1 - 2e_2)/\sqrt{6},$$
$$f_3 = (e_0 + e_1 + e_2 - 3e_3)/\sqrt{12}, \quad f_4 = (e_0 + e_1 + e_2 + e_3 - 4e_4)/\sqrt{20},$$

so that $\rho_{(4,1)}(\sigma)$ is orthogonal for all $\sigma \in S_5$.

2.4 The Diaconis–Shahshahani Upper Bound Lemma

Theorem 2.7 (Diaconis–Shahshahani [7])**.** *For any probability distribution Q on a finite group G,*

$$4 \cdot d_{\mathrm{TV}}(Q, U)^2 \leq \sum_{\rho \neq \mathrm{trivial}} d_\rho \cdot \|\widehat{Q}(\rho)\|_{\mathrm{F}}^2,$$

where the sum is over non-trivial irreducible representations ρ of dimension d_ρ, $\widehat{Q}(\rho) = \sum_{g \in G} Q(g)\rho(g)$ is the Fourier transform, and $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm.

For S_5, since the standard representation $(4,1)$ has the largest character value at transpositions ($\chi_{(4,1)}(\sigma)/d_{(4,1)} = 2/4 = 1/2$, compared to $\leq 1/5$ for other non-trivial representations), it is the *mixing bottleneck*. The remaining representations contract faster, so it suffices to bound $\|\widehat{D}(4,1)\|_{\mathrm{F}}^2$.

3 Exact T-Independence

Theorem 3.1 (T-Independence)**.** *Let* BP_T *be the Barrington branching program for the point function* CC_T *over* $\{0,1\}^\ell$*. Then* $D_T = D_{T'}$ *for all* $T, T' \in \{0,1\}^\ell$*.*

Proof. Let $S = \{i : T_i \neq T'_i\}$ be the set of differing bit positions. Define $F_S : \{0,1\}^L \to \{0,1\}^L$ by

$$F_S(b)_j = \begin{cases} 1 - b_j & \text{if } i_j \in S, \\ b_j & \text{otherwise,} \end{cases}$$

where i_j is the variable read at step j.

We verify three properties:

(1) F_S **is an involution.** Applying F_S twice flips each bit in S-positions twice, returning to the original string. Hence $F_S \circ F_S = \mathrm{Id}$.

(2) F_S **preserves inconsistency.** F_S flips *all* readings of each variable x_i with $i \in S$ simultaneously. If all readings of x_i were equal (say, all 0), after flipping they are all 1 — still consistent. If some readings differed, after flipping they still differ. Variables with $i \notin S$ are unaffected. Therefore inconsistency of each variable is preserved individually.

(3) $\varphi_{T'}(b) = \varphi_T(F_S(b))$ **for all** b**.** At each step j reading variable x_{i_j}:

- If $i_j \notin S$: $T_{i_j} = T'_{i_j}$ and F_S does not flip b_j. Same input, same target bit, same output.
- If $i_j \in S$: $T'_{i_j} = 1 - T_{i_j}$ and F_S flips b_j. The swap of target bit and input bit cancel: $g_j^{(b_j)}$ under T' equals $g_j^{(1-b_j)}$ under T, which equals $g_j^{(F_S(b)_j)}$ under T.

Hence the per-step outputs, and their product, are equal.

Combining (1)–(3):

$$\begin{aligned} D_{T'}(\pi) &= \Pr[\varphi_{T'}(b) = \pi \mid b \in \mathcal{I}] \\ &= \Pr[\varphi_T(F_S(b)) = \pi \mid b \in \mathcal{I}] && \text{(by (3))} \\ &= \Pr[\varphi_T(b') = \pi \mid F_S^{-1}(b') \in \mathcal{I}] && \text{(substituting } b' = F_S(b)) \\ &= \Pr[\varphi_T(b') = \pi \mid b' \in \mathcal{I}] && \text{(by (2), since } F_S = F_S^{-1} \text{ by (1))} \\ &= D_T(\pi). && \square \end{aligned}$$

Remark 3.2. *The proof uses no property of* S_5 *— it works for branching programs of any width over any group. It applies to all programs for point functions, not just Barrington's construction.*

Remark 3.3. *The proof extends to conjunctions with wildcards: if position* i *is a wildcard, the corresponding leaf outputs the same permutation* σ_i *for both bit values. Flipping* b_j *at wildcard positions changes both branches identically, so the output is unchanged. The bijection* F_S *is restricted to non-wildcard differing positions.*

Remark 3.4 (Unconditional T-independence)**.** *The bijection* F_S *preserves inconsistency of each variable individually (property (2)), and therefore also preserves the set of* consistent *paths. Consequently,* $D_T = D_{T'}$ *holds not only for the distribution conditioned on inconsistent paths, but also for the unconditional distribution over all* 2^L *paths and for the distribution over consistent paths alone. The conditioning on inconsistency is needed for the security argument (consistent paths reveal* T*), not for* T*-independence itself.*

Remark 3.5 (Variable-grouping invariance)**.** *The distribution* D_T *depends on the choice of transpositions assigned to each leaf, but* not *on how variables are grouped into the binary commutator tree. Exhaustive computation for* $\ell = 4$ *confirms that all three possible balanced groupings of* 4 *variables yield identical distributions. This is because* F_S *acts on variables, not on tree positions.*

4 Spectral Contraction

By Theorem 3.1, D_T is independent of T, so it suffices to show $d_{\mathrm{TV}}(D_T, U) \leq \mathrm{negl}(\ell)$ for any fixed T. By the Diaconis–Shahshahani bound (Theorem 2.7), this reduces to bounding $\|\widehat{D}(\rho)\|_{\mathrm{F}}^2$ for each non-trivial representation ρ.

Remark 4.1 (Sign representation vanishes identically)**.** *For the sign representation* (1^5)*, the leaf Fourier transform is* $\widehat{D}(\mathrm{sign}) = \frac{1}{2}(\mathrm{sign}(\sigma) + 1) = 0$ *for any transposition* σ*, since* $\mathrm{sign}(\sigma) = -1$*. This propagates to all levels by multiplicativity, so the sign representation contributes nothing to the Diaconis–Shahshahani bound. The sum thus runs over* 5 *(not* 6*) non-trivial representations.*

We focus on the bottleneck representation $(4, 1)$.

4.1 Framework: Projections and the Fourier Domain

For a transposition $\sigma = (a\ b) \in S_5$, the standard representation $\rho_{(4,1)}(\sigma)$ has eigenvalues $\{+1, +1, +1, -1\}$ (three fixed directions and one flipped). The Fourier transform of the leaf distribution $\{\sigma : 1/2,\ e : 1/2\}$ is

$$P_\sigma := \frac{1}{2}\big(\rho_{(4,1)}(\sigma) + I_4\big),$$

which is a rank-3 orthogonal projection (idempotent, symmetric, trace 3).

Lemma 4.2. *For any transposition* $\sigma \in S_5$*:*
(a) $P_\sigma^2 = P_\sigma$ *(idempotent),*
(b) $P_\sigma^\top = P_\sigma$ *(symmetric),*
(c) $\mathrm{rank}(P_\sigma) = 3$*,*
(d) $\|P_\sigma\|_{\mathrm{F}}^2 = \mathrm{tr}(P_\sigma) = 3$*.*

The null space of P_σ is spanned by the projection of $e_a - e_b$ onto the hyperplane V, where $\sigma = (a\ b)$.

At each level of the Barrington commutator tree, the Fourier transform is a product of projections. For the single AND block at level 1 using transpositions σ, τ (with $\sigma^{-1} = \sigma$, $\tau^{-1} = \tau$ since they are transpositions):

$$\widehat{D}_1 = P_\sigma \cdot P_\tau \cdot P_\sigma \cdot P_\tau = (P_\sigma P_\tau)^2.$$

At level 2, using two level-1 blocks A (with σ_1, τ_1) and B (with σ_2, τ_2):

$$\widehat{D}_2 = (P_{\sigma_1} P_{\tau_1})^2 \cdot (P_{\sigma_2} P_{\tau_2})^2 \cdot (P_{\tau_1} P_{\sigma_1})^2 \cdot (P_{\tau_2} P_{\sigma_2})^2,$$

where the inverse blocks reverse the transposition order.

4.2 Theorem 1: Level 0 to Level 1 Contraction

Theorem 4.3 (Universal Level-1 Contraction)**.** *For any pair of adjacent transpositions* (σ, τ) *in* S_5 *(i.e.,* $[\sigma, \tau] \neq e$*):*

$$\|(P_\sigma P_\tau)^2\|_{\mathrm{F}}^2 = \frac{129}{64}.$$

The contraction ratio is $\frac{129/64}{3} = \frac{43}{64} \approx 0.6719$*, and is* ***universal*** *— identical for all* 30 *adjacent transposition pairs.*

Proof. There are $\binom{5}{2} = 10$ transpositions in S_5, forming 30 adjacent pairs (pairs sharing exactly one element, equivalently $[\sigma, \tau] \neq e$). For each pair, we compute $(P_\sigma P_\tau)^2$ in the orthonormal basis of Section 2 using exact rational arithmetic, and verify $\|(P_\sigma P_\tau)^2\|_{\mathrm{F}}^2 = 129/64$.

The universality follows from the geometry of the standard representation: the null vectors of P_σ and P_τ (projections of $e_a - e_b$ and $e_c - e_d$ onto V) have inner product $\pm 1/2$ for all adjacent pairs, determined by the simplex geometry of $\{e_0, \ldots, e_4\}$ in V. Since $\|(P_\sigma P_\tau)^2\|_{\mathrm{F}}^2$ depends only on this angle, it is universal. □

4.3 Theorem 2: Level 1 to Level 2 Contraction

Theorem 4.4 (Level-2 Contraction)**.** *For any valid quadruple of transpositions* $(\sigma_1, \tau_1, \sigma_2, \tau_2)$ *with* $[\sigma_i, \tau_i] \neq e$ *and* $[[\sigma_1, \tau_1], [\sigma_2, \tau_2]] \neq e$*:*

$$\|\widehat{D}_2\|_{\mathrm{F}}^2 \leq \frac{67586345}{67108864} \approx 1.0071.$$

The contraction ratio relative to level 1 *is at most* $\frac{67586345/67108864}{129/64} \approx 0.4997 < 1$.

Proof. We enumerate all 405 valid quadruples (from the 30 adjacent pairs with the additional constraint that the outer commutator is nontrivial). For each quadruple, we compute $\widehat{D}_2$ in exact rational arithmetic and evaluate $\|\widehat{D}_2\|_{\mathrm{F}}^2$. The worst case over all 405 quadruples is 67586345/67108864. The computation yields 16 distinct Frobenius norms, ranging from $82431849/1073741824 \approx 0.077$ to the worst case above. □

4.4 The Simplex Obstruction

Despite the Frobenius-norm contraction at level 2, the operator norm $\|\widehat{D}_2\|_{\mathrm{op}}$ equals 1 for 270 of the 405 valid quadruples, blocking a naive submultiplicativity argument for higher levels. The following observation explains the structure of this obstruction.

Observation 4.5 (Simplex Structure)**.** *The* 270 *quadruples with* $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ *each preserve exactly one direction in* $\mathbb{R}^4$ *(as the eigenvector of* $\widehat{D}_2^\top \widehat{D}_2$ *with eigenvalue* 1*). These directions are precisely the projections of the* 5 *standard basis vectors* $e_0, \ldots, e_4 \in \mathbb{R}^5$ *onto the hyperplane* $V = \{\sum v_i = 0\}$*:*

$$v_k = \frac{e_k - \bar{e}}{\|e_k - \bar{e}\|}, \quad \bar{e} = \tfrac{1}{5}(1,1,1,1,1)^\top, \quad k = 0, \ldots, 4.$$

These 5 *directions form a* ***regular simplex*** *in* $\mathbb{R}^4$*, with pairwise inner products* $|\langle v_i, v_j \rangle| = 1/4$ *for all* $i \neq j$*. Each direction is preserved by exactly* 54 *quadruples (perfectly symmetric).*

The regularity of this structure — five directions, uniform distribution, constant pairwise angle — suggests a general phenomenon rather than an artifact of S_5.

Conjecture 4.6 (Simplex universality for S_n)**.** *For* S_n *with* $n \geq 5$*, the level-*2 *quadruples with operator norm* 1 *in the standard representation* $(n-1, 1)$ *preserve directions that are the projections of the* n *standard basis vectors onto the hyperplane* $V = \{v \in \mathbb{R}^n : \sum v_i = 0\}$*, forming a regular simplex in* $\mathbb{R}^{n-1}$ *with pairwise* $|\langle v_i, v_j \rangle| = 1/(n-1)$.

If true, the direction-mixing mechanism at level 3 would generalize to branching programs of arbitrary width, since the commutator constraint $[\alpha_A, \alpha_B] \neq e$ forces incompatible directions regardless of n. This is relevant because width > 5 programs appear in optimized obfuscation constructions that avoid Barrington's 4^d blowup [2].

4.5 Theorem 3: Level 2 to Level 3 Contraction

Lemma 4.7 (Direction Mixing)**.** *For every valid level-3 configuration* $(\mathrm{quad}_A, \mathrm{quad}_B)$ *in which both sub-blocks have* $\|\widehat{D}_2\|_{\mathrm{op}} = 1$*, the two sub-blocks preserve* ***different*** *directions. Specifically, of the* 29160 *such pairs, zero share the same preserved direction.*

Proof. Exhaustive enumeration. For each of the 73810 valid level-3 pairs (those with nontrivial outer commutator $[\alpha_A, \alpha_B] \neq e$ where α_A, α_B are the level-2 targets), we check the preserved directions. Among the 29160 pairs where both sub-blocks are "problematic" (operator norm = 1), zero share the same direction. □

The intuition is that a quadruple preserving direction v_k uses transpositions that fix element k. For the outer commutator to be nontrivial, the second quadruple must move element k, necessarily changing the preserved direction.

Theorem 4.8 (Level-3 Contraction)**.** *For every valid level-3 configuration:*

$$\|\widehat{D}_3\|_{\mathrm{op}} \leq \lambda := 0.0354.$$

In particular, $\|\widehat{D}_3\|_{\mathrm{op}} < 1$ *for all* 73810 *configurations.*

Proof. For each valid level-3 pair $(\mathrm{quad}_A, \mathrm{quad}_B)$, we compute

$$\widehat{D}_3 = \widehat{D}_{2,A} \cdot \widehat{D}_{2,B} \cdot \widehat{D}_{2,A}^\top \cdot \widehat{D}_{2,B}^\top$$

and evaluate the operator norm (maximum singular value) numerically. The maximum over all 73810 configurations is $\lambda \approx 0.0353$, strictly less than 1.

The contraction occurs because:

- If at least one sub-block has $\|\widehat{D}_2\|_{\mathrm{op}} < 1$, submultiplicativity gives $\|\widehat{D}_3\|_{\mathrm{op}} \leq \|\widehat{D}_{2,A}\|_{\mathrm{op}}^2 \cdot \|\widehat{D}_{2,B}\|_{\mathrm{op}}^2 < 1$.
- If both sub-blocks have $\|\widehat{D}_2\|_{\mathrm{op}} = 1$ but preserve different directions $v_A \neq v_B$ (guaranteed by Theorem 4.7), then $\widehat{D}_{2,A}$ maps v_B to a vector with component $|\langle v_A, v_B\rangle| = 1/4$ along v_A, causing the product to contract. □

4.6 Theorem 4: Induction to All Levels

Theorem 4.9 (Exponential Contraction)**.** *For all Barrington branching programs for point functions* CC_T *over* $\{0,1\}^\ell$ *with* $\ell = 2^d$*,* $d \geq 3$*:*

$$\|\widehat{D}_d\big((4,1)\big)\|_{\mathrm{op}} \leq \lambda^{4^{d-3}},$$

where $\lambda = 0.0354$*.*

Proof. At depth $d+1$, the commutator structure gives $\widehat{D}_{d+1} = \widehat{D}_A \cdot \widehat{D}_B \cdot \widehat{D}_A^\top \cdot \widehat{D}_B^\top$. By submultiplicativity of the operator norm and $\|M^\top\|_{\mathrm{op}} = \|M\|_{\mathrm{op}}$:

$$\|\widehat{D}_{d+1}\|_{\mathrm{op}} \leq \|\widehat{D}_A\|_{\mathrm{op}} \cdot \|\widehat{D}_B\|_{\mathrm{op}} \cdot \|\widehat{D}_A^\top\|_{\mathrm{op}} \cdot \|\widehat{D}_B^\top\|_{\mathrm{op}} = \|\widehat{D}_d\|_{\mathrm{op}}^4.$$

The base case $\|\widehat{D}_3\|_{\mathrm{op}} \leq \lambda$ is Theorem 4.8. Iterating: $\|\widehat{D}_d\|_{\mathrm{op}} \leq \lambda^{4^{d-3}}$. □

5 Main Result

Theorem 5.1 (Main Theorem)**.** *For all Barrington branching programs for point functions* CC_T *over* $\{0,1\}^\ell$ *with* $\ell = 2^d$*,* $d \geq 3$*:*

(a) $D_T = D_{T'}$ *for all* $T, T' \in \{0,1\}^\ell$*.*

(b) $d_{\mathrm{TV}}(D_T, \mathrm{Uniform}(S_5)) \leq 2 \cdot \lambda^{4^{d-3}}$*, where* $\lambda = 0.0354$*.*

(c) In particular, $d_{\mathrm{TV}}(D_T, U) = \mathrm{negl}(\ell)$*.*

Proof. Part (a) is Theorem 3.1. For part (b), by Theorem 2.7:

$$d_{\mathrm{TV}}(D_T, U)^2 \leq \frac{1}{4} \sum_{\rho \neq \mathrm{triv}} d_\rho \cdot \|\widehat{D}(\rho)\|_{\mathrm{F}}^2.$$

The dominant term is $d_{(4,1)} \cdot \|\widehat{D}(4,1)\|_{\mathrm{F}}^2 = 4 \cdot \|\widehat{D}(4,1)\|_{\mathrm{F}}^2$. Since $\|M\|_{\mathrm{F}}^2 \leq d \cdot \|M\|_{\mathrm{op}}^2$ for a $d \times d$ matrix:

$$\|\widehat{D}(4,1)\|_{\mathrm{F}}^2 \leq 4 \cdot \|\widehat{D}(4,1)\|_{\mathrm{op}}^2 \leq 4\lambda^{2 \cdot 4^{d-3}}.$$

Therefore $d_{\mathrm{TV}}(D_T, U)^2 \leq 4 \cdot \lambda^{2 \cdot 4^{d-3}}$, giving $d_{\mathrm{TV}}(D_T, U) \leq 2 \cdot \lambda^{4^{d-3}}$. For part (c), since $\lambda < 1$ and 4^{d-3} grows super-exponentially in $d = \log_2 \ell$, the bound is negligible in ℓ. □

For concrete values:

ℓ	Depth d	$\|\widehat{D}\|_{\mathrm{op}}$ bound	d_{TV} bound
8	3	3.54×10^{-2}	7.07×10^{-2}
16	4	1.55×10^{-6}	3.11×10^{-6}
32	5	5.83×10^{-24}	1.17×10^{-23}
64	6	1.15×10^{-93}	2.30×10^{-93}

6 Extensions and Open Problems

6.1 Conjunctions with Wildcards

A conjunction $\mathrm{CC}_{T,W}$ with target $T \in \{0,1\}^\ell$ and wildcard set $W \subseteq [\ell]$ outputs 1 iff $x_i = T_i$ for all $i \notin W$. In the branching program, wildcard positions output σ_i for both bit values. Obfuscation of conjunctions has been studied extensively [4, 5, 6]; our results provide an information-theoretic complement to the computational security guarantees in that literature.

Theorem 3.1 extends directly to conjunctions: $D_{T,W} = D_{T',W}$ for all T, T' with the same wildcard set W. The spectral contraction depends on the wildcard density $|W|/\ell$. Computational evidence shows strong contraction for $|W|/\ell \leq 1/2$, weakening contraction for $|W|/\ell = 3/4$, and no contraction for $|W|/\ell = 1$ (the trivial function). The precise threshold is an open problem.

6.2 Beyond Point Functions

For general branching programs computing the same Boolean function via different constructions, "strong universal mixing" ($D_{C_0} = D_{C_1}$) is **false**: programs with different transposition assignments produce different distributions. However, "weak universal mixing" (both distributions converge to uniform independently) appears to hold: computational evidence shows equivalent circuits with different pair assignments achieving similar d_{TV} values. Proving this would strengthen the security guarantee for indistinguishability obfuscation of general NC^1 functions.

Computational experiments reveal a stronger pattern. For $\ell = 4$, two programs computing AND_4 with different transposition pairs yield Fourier norms $\|\widehat{D}(4,1)\|_{\mathrm{F}}^2 = 0.147$ and 0.293 respectively — clearly different distributions — yet *identical* total variation distances $d_{\mathrm{TV}} = 0.309$ to uniform. The same equidistance persists for OR-of-AND circuits ($d_{\mathrm{TV}} \approx 0.19$ for two structurally different implementations). If this "equidistance" phenomenon is general, it would mean that $d_{\mathrm{TV}}(D_C, U)$ depends only on the Boolean function computed, not on the specific Barrington construction — a property strictly stronger than weak universal mixing.

6.3 Generalization to Other Groups

Our Fourier framework is specific to S_5. The branching programs in modern lattice-based iO constructions [3, 19, 13] operate over matrix groups $\mathrm{GL}_n(\mathbb{Z}_q)$. Extending the spectral contraction analysis to these groups would directly address the security of post-quantum iO candidates, but requires handling infinite-dimensional representation theory. An intermediate step would be establishing analogous results for S_6 and S_7, testing whether the simplex structure and direction-mixing phenomenon generalize to larger symmetric groups.

Our proof follows a three-step program that is, in principle, portable: (1) define the output distribution on inconsistent inputs, (2) identify the bottleneck irreducible representation, (3) discover geometric structure in the spectral obstructions. For $\mathrm{GL}_2(\mathbb{F}_q)$ with q a small prime, the irreducible representations are completely classified (principal series, cuspidal, Steinberg, and one-dimensional characters), and the analogue of step (2) is computationally feasible. Determining whether the level-2 obstructions for matrix branching programs over $\mathrm{GL}_2(\mathbb{F}_q)$ exhibit geometric regularity analogous to the simplex in Theorem 4.5 would be a concrete first test of whether spectral contraction extends beyond symmetric groups.

If such structure exists, it could provide *unconditional* mixing guarantees for matrix branching programs — replacing cryptographic assumptions (e.g., evasive LWE [19]) with algebraic theorems in specific parameter regimes. This prospect is especially timely given recent impossibility results for broad families of LWE-based assumptions [9].

6.4 Algebraic Proof of Direction Mixing

Theorem 4.7 is proved by exhaustive enumeration. An algebraic proof explaining *why* valid level-3 configurations cannot share preserved directions would strengthen the result and potentially generalize to arbitrary depth without case analysis. The conjectured mechanism is that a quadruple preserving direction v_k (the projection of e_k) consists exclusively of transpositions fixing element k, while a nontrivial outer commutator requires the second quadruple to move element k. Formalizing this would reduce the exhaustive computation to a short group-theoretic argument.

7 Computational Verification

The proofs of Theorems 4.3 and 4.4 use exact rational arithmetic (implemented in `sympy`) with a verified orthonormal basis for the standard representation. Theorem 4.8 uses numerical computation (`numpy`) over all 73810 valid configurations.

The full verification suite consists of 12 scripts, publicly available at:

https://github.com/edpz/barrington-mixing

Script	Content	Method
01	Ambivalence of S_5 (similarity blindness)	Exhaustive
02	Collision analysis, $\ell = 4$	Exhaustive
03	Collision scaling, $\ell = 4 \to 8$	Sampled
04	Fourier spectral analysis	Exact + sampled
05	Barrington construction + Kilian obfuscation	Exhaustive
06	Contraction operator (unconditional distribution)	Analytic
07	T-independence proof verification	Exhaustive
08	Conjunctions with wildcards	Exhaustive
09	Universal mixing investigation	Exact + sampled
10	Complete spectral contraction proof	Exact + numerical
11	Eigenvector analysis (simplex discovery)	Numerical
12	Level-3 exhaustive verification	Numerical

Scripts 10–12 collectively verify the main theorem in under 15 seconds on a standard laptop (the exact-arithmetic steps in Script 10 dominate). Scripts 01–09 provide additional cross-validation and extensions.

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